

Discrete Structure Lab-sheet Questions

1. WAP in python to display truth table of negation.
2. WAP in python to display truth table of conjunction.
3. WAP in python to display truth table of disjunction.
4. WAP in python to display truth table of implication.
5. WAP in python to display truth table of bi-implication.
6. WAP in python to verify tautology or not.
7. WAP in python to find the GCD between two input numbers using Euclidean algorithm.
8. WAP in python to find the GCD between two input numbers using Extended Euclidean algorithm.
9. WAP in python to solve the given simultaneous congruence using Chinese Remainder theorem.
10. WAP in python to test relation properties: Reflexive, Symmetric, Anti-symmetric and Transitive for a given set and Relation.
11. WAP in python to represent relation in matrix form for a given set and relation.
12. WAP in python to represent relation in digraph form for a given set and relation.
13. WAP in python to determine whether the given relation is equivalence relation or not.
14. WAP in python to determine whether the given relation is partial order relation or not.
15. WAP in python to find the factorial of an input number using recursion.
16. WAP in python to find the n^{th} term of Fibonacci sequence using recursion.
17. WAP in python to validate inductive proof.
18. WAP in python to represent adjacency list and visualize using NetworkX.
19. WAP In python to represent adjacency matrix and visualize using NetworkX.
20. WAP in python to perform tree traversal (pre-order, in-order and post-order).
21. WAP in python to find MST using Kruskal's algorithm.
22. WAP in python to construct finite automata that accepts strings only **ab** and **abb**.

The format for each lab question report is as follows:

Theory

------(Write definition, truth table or mathematical formula depending on question here)-----

Source Code

(Write program code in this section)

Output:

------(Screenshot)-----

Conclusion

-----**(write in 1-2 line concluding what have you done, how did you do, what did you learn.....)**.....

Note: Please prepare a lab report in A4 size paper format with only one side written.